

[NJC/2024/Q2]

2 Name your Jupyter Notebook as:

TASK2_<your name>.ipynb

A linked list is a collection of data elements, whose order is not given by their physical placement in memory. Instead, each element points to the next.

For each of the sub-tasks, add a comment statement at the beginning of the code, using the hash symbol '#' to indicate the sub-task the program code belongs to, for example:

```
In [1]: #Task 2.1  
Program Code
```

Output:

Task 2.1

Write the LinkedList class in Python. Use of a simple Python list is not sufficient. Include the following methods:

- insert(data) inserts the data at the beginning (head) of the list
- delete(data) attempts to delete data from the list; if the item was not present, return None
- search(data) returns a Boolean value: True if data is in the list, False if not in the list
- count() should return the number of elements in the list, or zero if empty
- to_String() should return a string containing a suitably formatted list with the elements separated by a comma and a space, with square brackets at either end, e.g. in the form:
[11, 2, 7, 4]

[8]

Test LinkedList by using the integers in the file TASK2DATA.TXT. Use the to_String() method to print the resulting contents of the list.

[3]

Task 2.2

In mathematics, a polynomial in x is an expression $p(x)$ consisting a single variable x and coefficients, that involves on the operations of addition, subtraction, multiplication and non-negative integer exponentiation of x .

For example, $x^2 - 4x + 7$, $8x^3 - 5x + 6$, x^5 , 5 are polynomials in x .

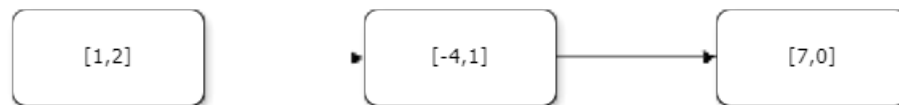
More generally, $p(x)$ is a polynomial in x if

$$p(x) = a_0x^0 + a_1x^1 + a_2x^2 + \cdots + a_nx^n,$$

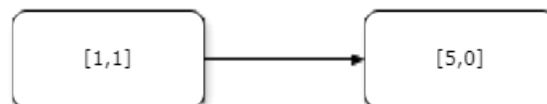
where $a_0, \cdots, a_n$ are the called coefficients of the term $x^0, \cdots, x^n$ respectively. The powers of x are called the exponents.

We can represent a polynomial in x as a linked list of nodes where each nodes contains 2 pieces of data, the coefficient and exponent.

Example 1, $x^2 - 4x + 7$ can be represented as:



Example 2, $x + 5$ can be represented as:



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Write a Python subclass Polynomial using LinkedList as its superclass.

The create method in the Polynomial subclass takes in a list of data items, each data item contains a coefficient and an exponent. The method should ensure that the elements are stored in descending order of the exponent.

The to_String method in the Polynomial subclass should return a string that represents the polynomial in descending power of the exponent. Coefficients of the polynomials should be enclosed within parentheses characters ().

Example, the method should return the linked list representation of the polynomials:

- $x^2 + x^3 - 5$ as "(1)*x**3 +(1)*x**2 + (-5)*x**0"
- $1 - x^4 + 2x$ as "(-1)*x**4 + (2)**x**1 + (1)*x**0" [6]

Test your program by creating Polynomial objects that represent the following polynomials:

- $p(x) = x^2 + x^3 - 5,$
- $q(x) = 1 - x^4 + 2x,$

Use the to_String() method to print the string representation of each of the polynomial. [2]

Task 2.3

Let $p(x)$ and $q(x)$ be polynomials such that

$$p(x) = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

$$q(x) = b_0 + b_1x + b_2x^2 + \cdots + b_mx^m$$

where m is not necessarily the same as n .

We define the addition of two polynomials $p(x) + q(x)$ to be

$$p(x) + q(x) = \begin{cases} (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_m + b_m)x^m + a_{m+1}x^{m+1} + \cdots + a_nx^n & \text{if } m \leq n, \\ (a_0 + b_0) + (a_1 + b_1)x + \cdots + (a_n + b_n)x^n + a_{n+1}x^{n+1} + \cdots + a_mx^m & \text{if } n < m \end{cases}$$

In other words, the resulting polynomial when two polynomials are added is a polynomial where the coefficients are the sum of the coefficients of $p(x)$ and $q(x)$ where the x term has the same exponent.

Amend your code for Task 2.2 to declare the Polynomial method `addPoly()` that takes a Polynomial object and returns another Polynomial object which represents the addition of the two polynomials. [7]

Test your `addPoly()` method on the Polynomial objects you created in Task 2.2 and use the `toString()` method to print the string representation of the polynomial. [1]

Save your Jupyter Notebook for Task 2.